Bonus: Colliders, selection bias, and loss to follow-up

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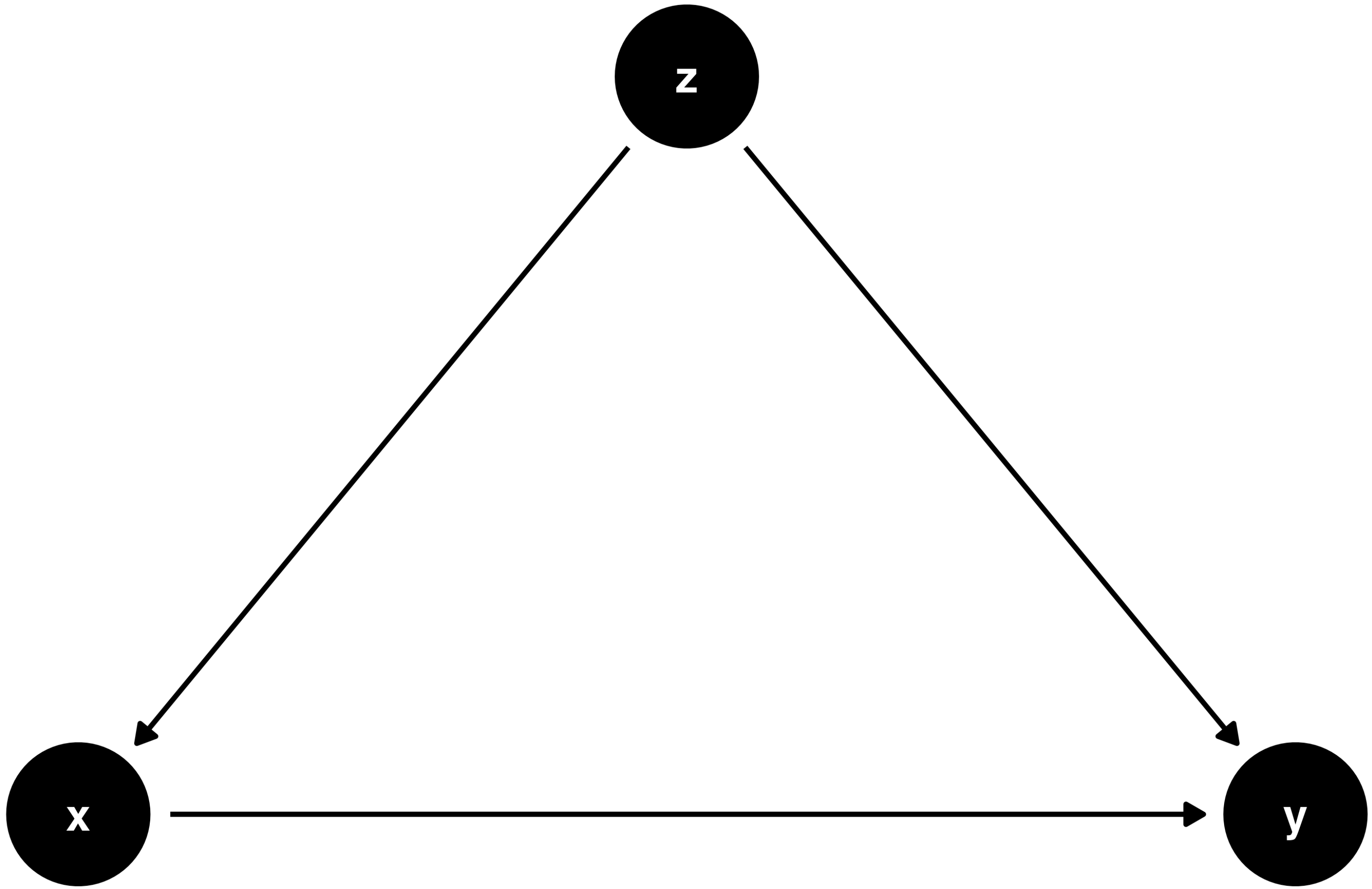
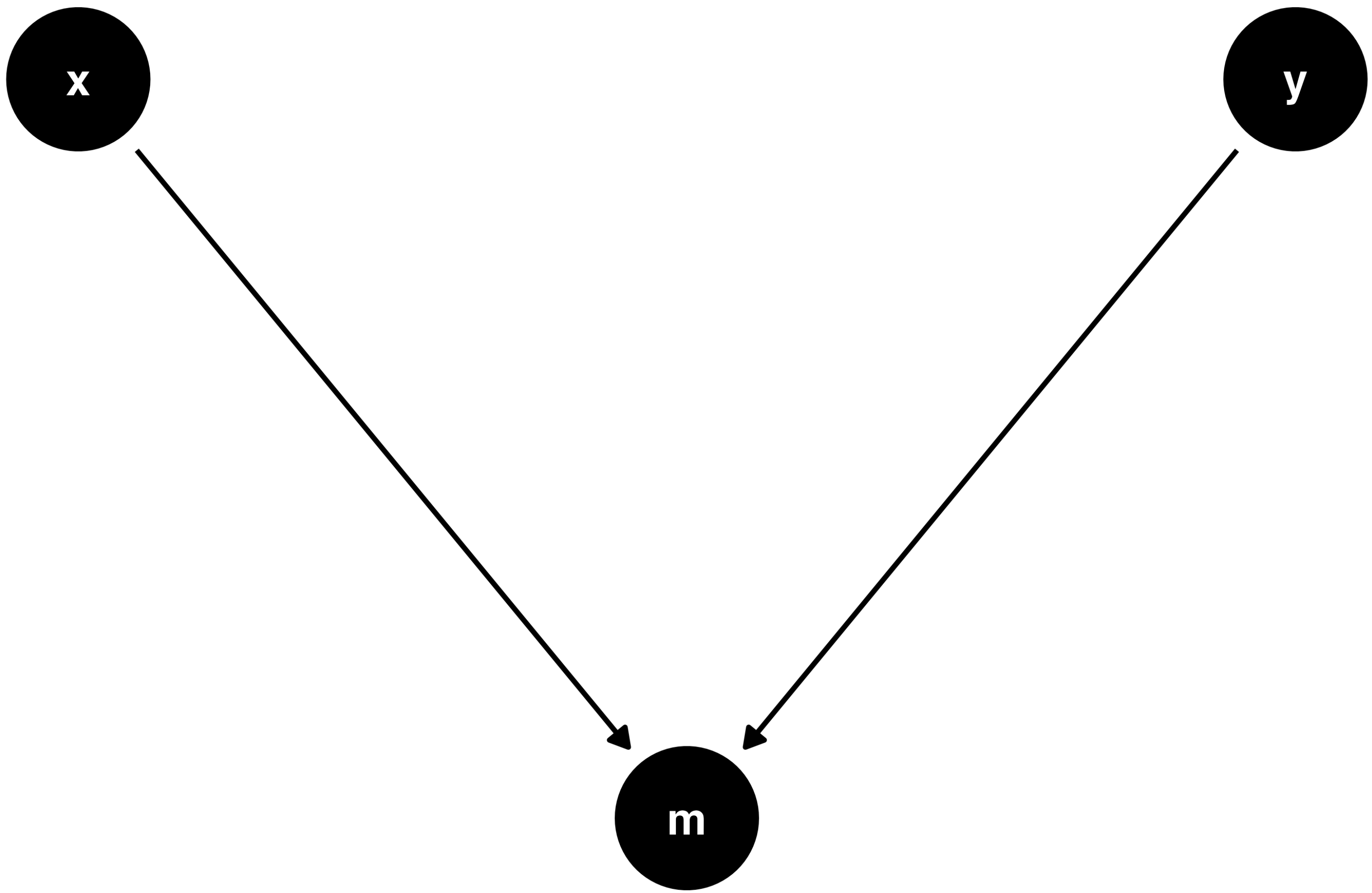
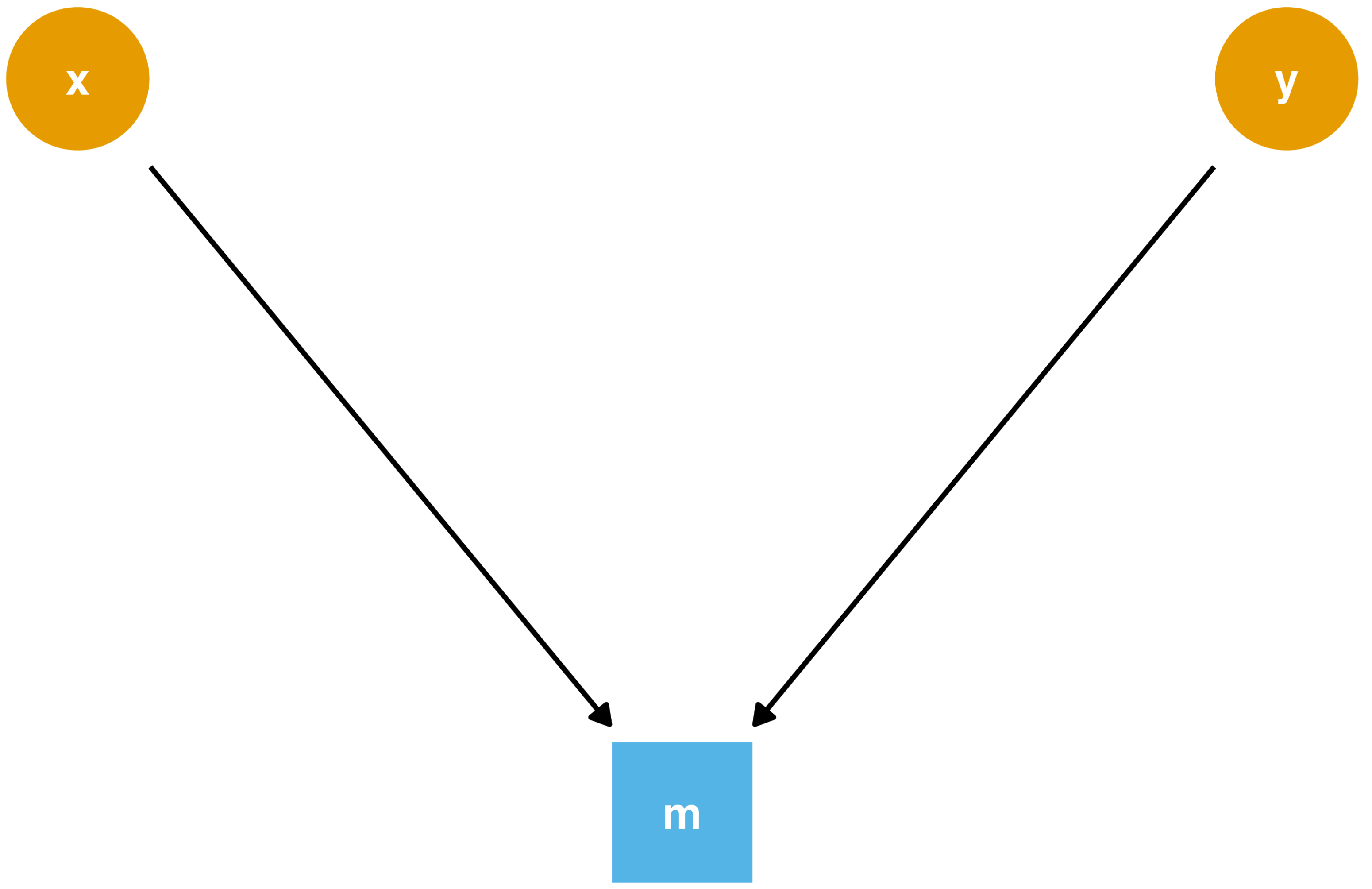
Confounders and chains



Colliders



Colliders



Let's prove it!

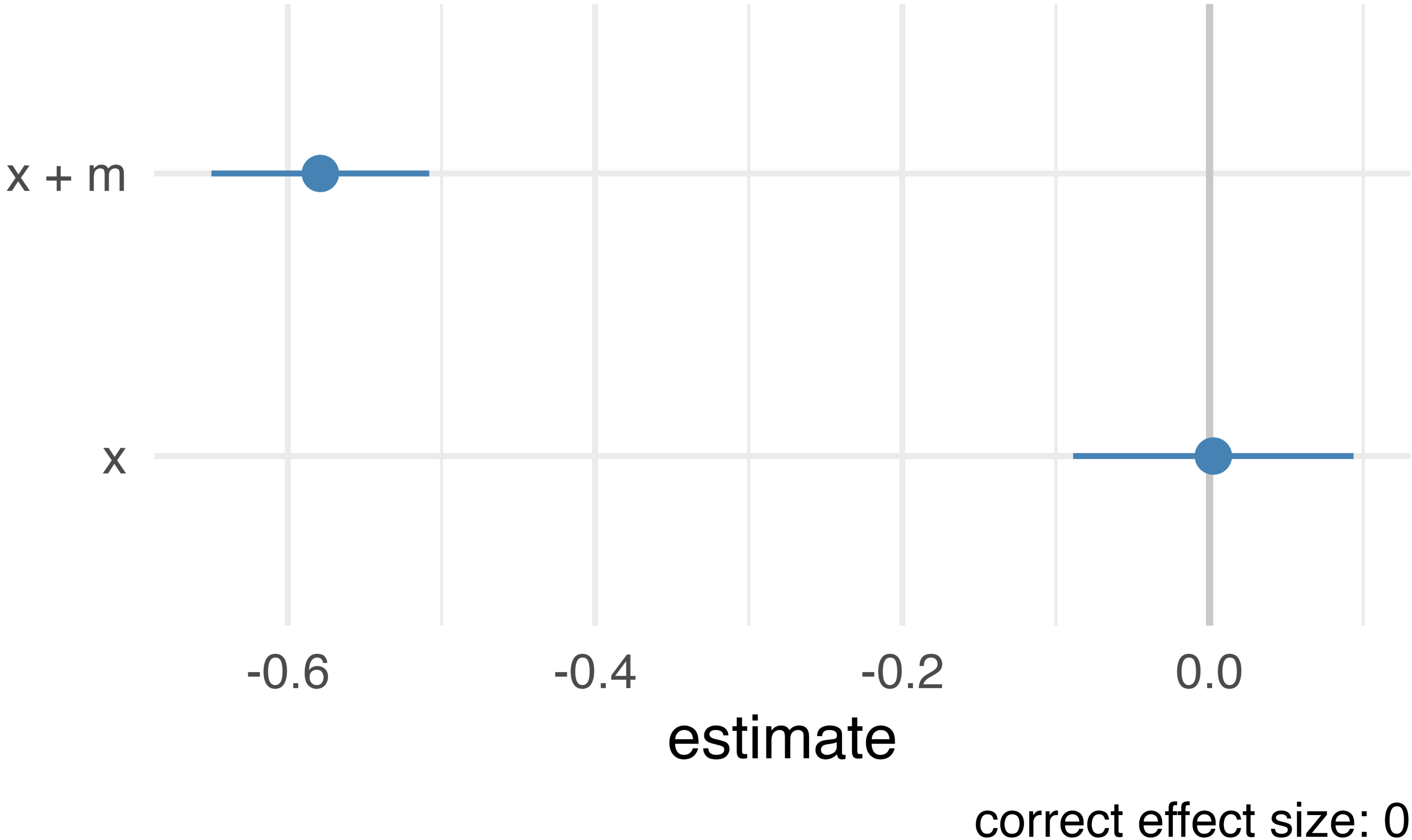
```r
set.seed(8)
collider_data <- collider_triangle() |>
  simulate_data(-0.6)
```

Let’s prove it!

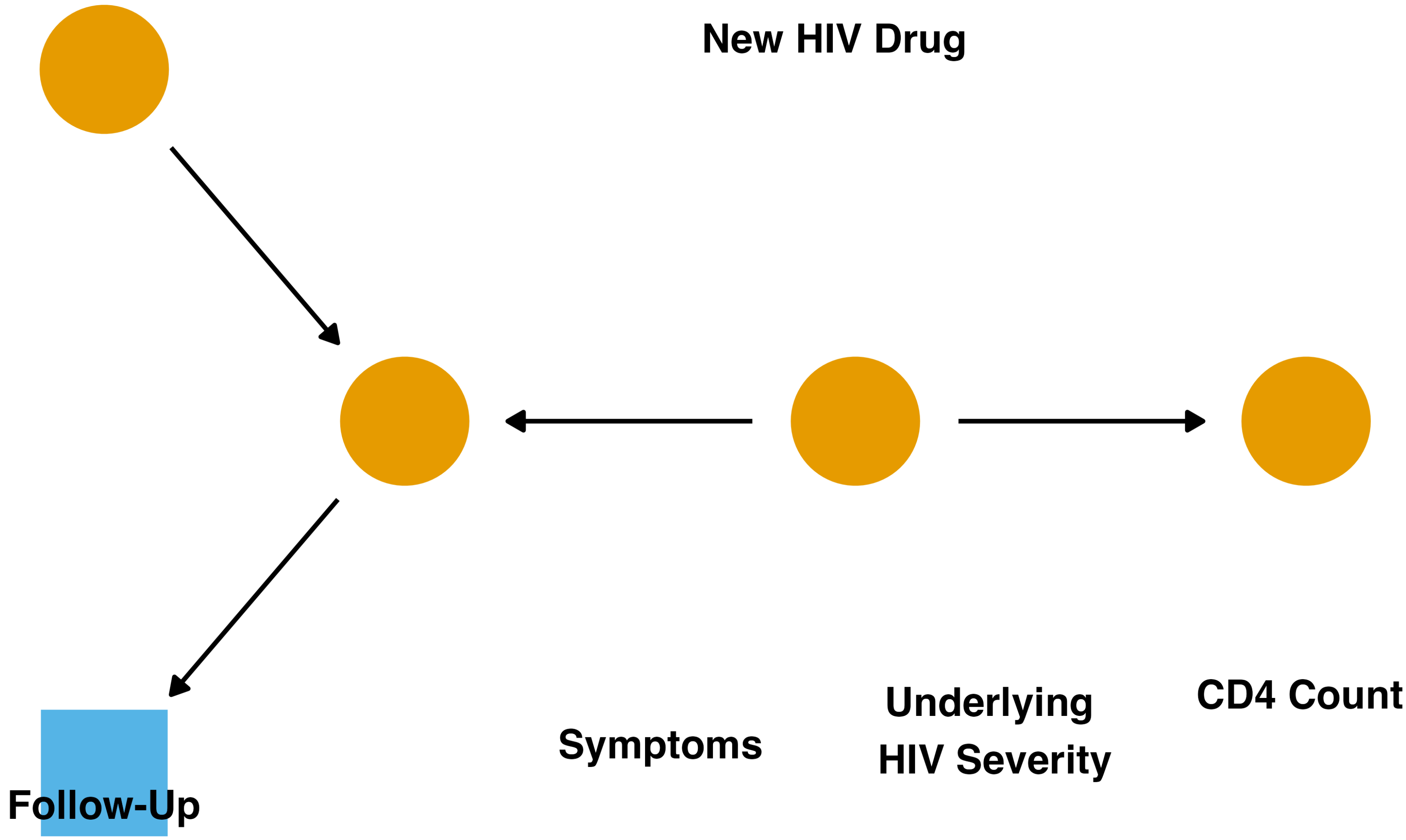
```r
collider_data
```

```
# A tibble: 500 × 3
         m       x       y
     <dbl>   <dbl>   <dbl>
 1 -0.366   0.174  -0.461
 2  1.21   -0.307  -0.273
 3 -0.441   0.0858  0.551
 4 -0.385   0.942   0.0108
 5  0.431  -0.205  -1.19
 6  0.172   0.990  -0.453
 7 -0.230   0.847  -0.710
 8 -0.769   2.08   -0.208
 9 -2.91    0.516   3.55
10 -0.897   0.680  -0.335
# ℹ 490 more rows
```

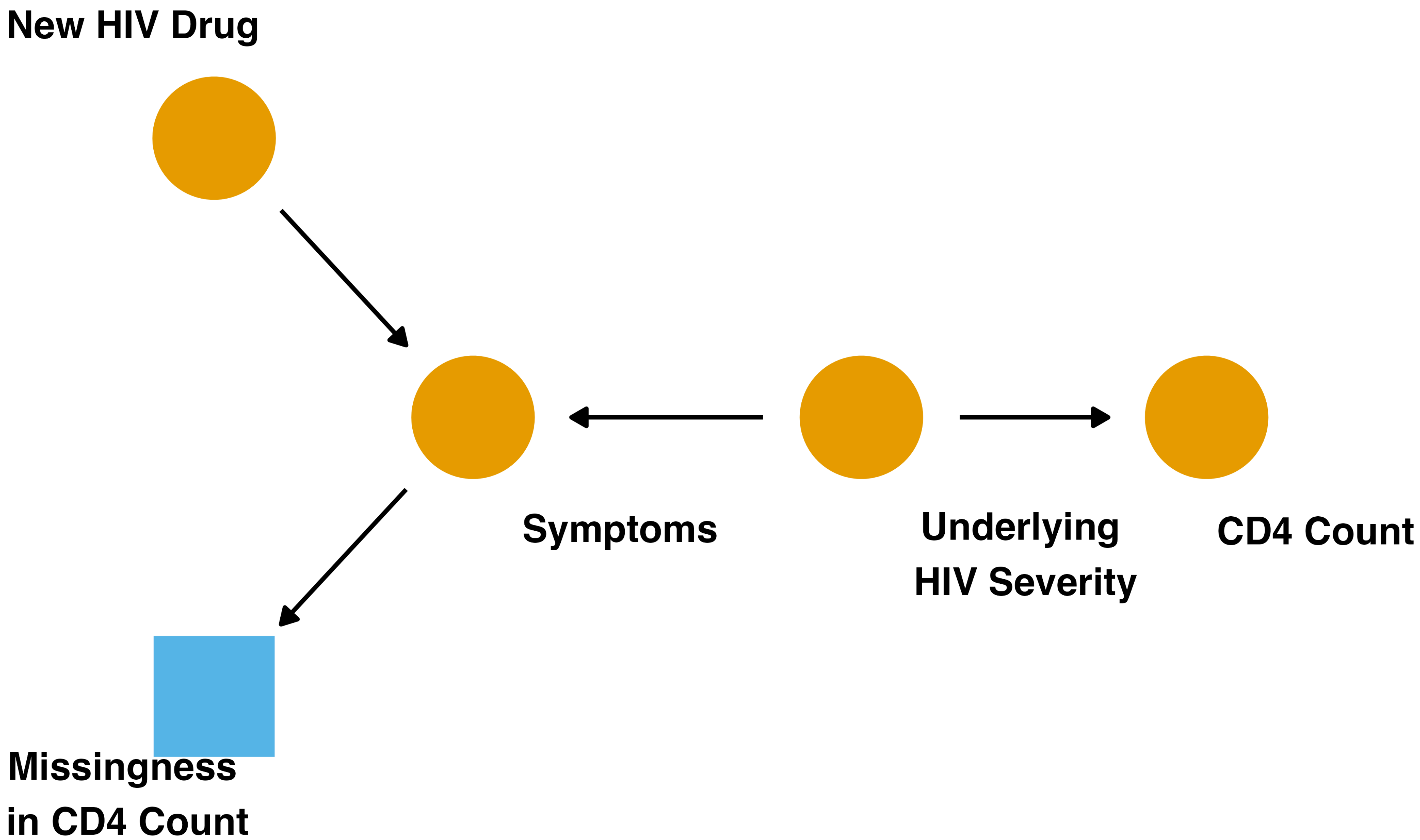
Let’s prove it!



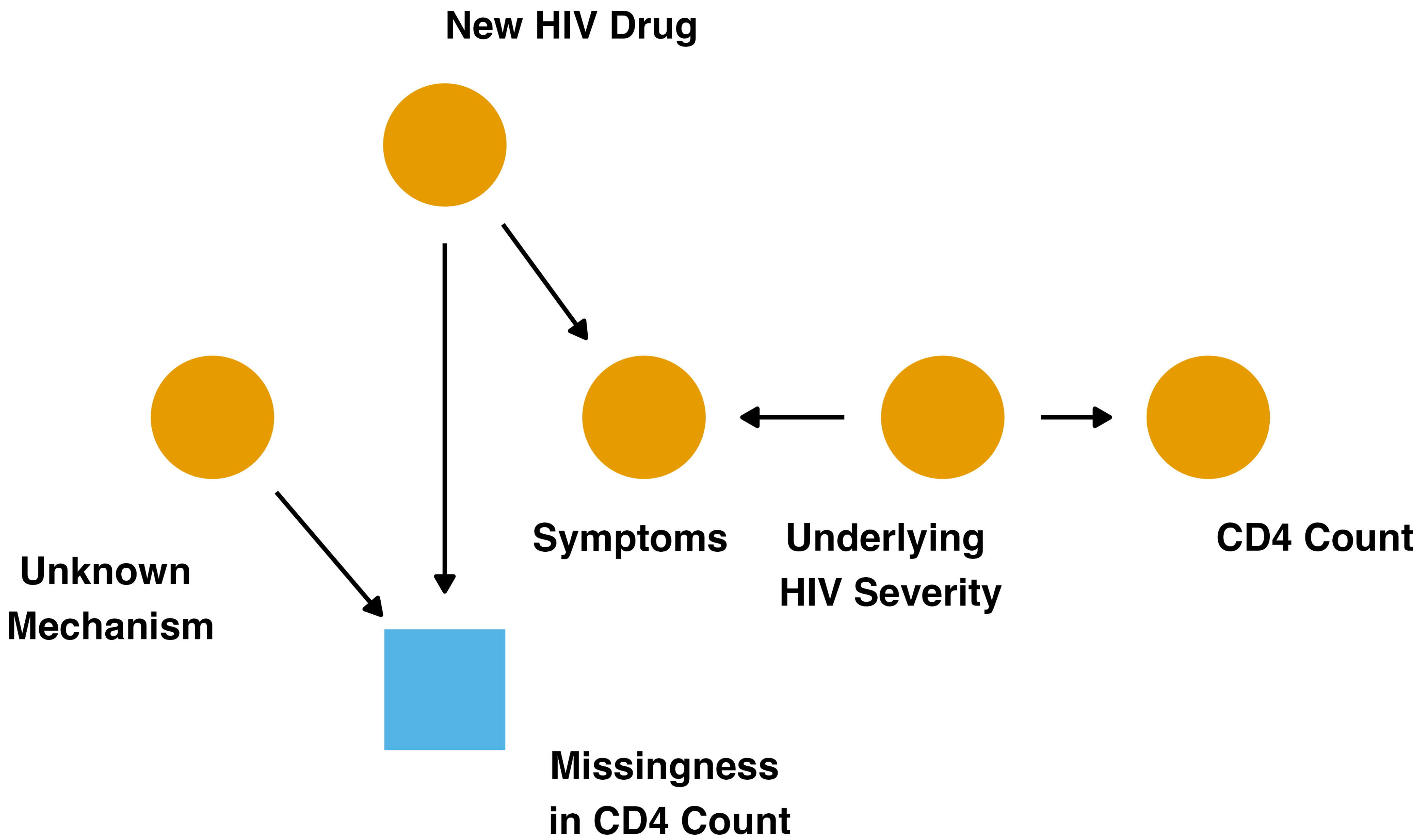
Loss to follow-up



Censoring as missingness



When censoring doesn't bias



What can we recover?

- We are always conditioning on the data we actually have
- When that opens no backdoor path, complete-case analysis still recovers the causal effect, with less precision
- We may still be unable to estimate other statistics, like the mean of the outcome
- When it does open a backdoor path, sometimes we can close it and sometimes we can’t
- Not all censoring needs weights

Change the estimand

- Sometimes the variable you select on is both a collider and a mediator

- Conditioning on it induces relationships between other variables, and it blocks the indirect effect

- We can't calculate the total effect, but we can still calculate the direct effect by adjusting for the selection variable

Adjusting for selection bias

1. Fit a probability of censoring model, e.g. *glm(censoring ~ predictors, family = binomial())*
2. Create weights using inverse probability strategy
3. Use weights in your causal model

We won't do it here, but you can include many types of weights in a given model. Just take their product, e.g. *multiply inverse propensity of treatment weights by inverse propensity of censoring weights*.

Your Turn

Work through Your Turns 1-3 in `16-bonus-selection-bias-exercises.qmd`

10:00